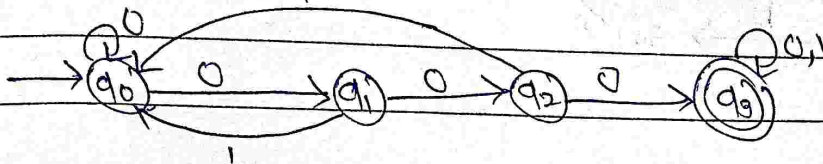
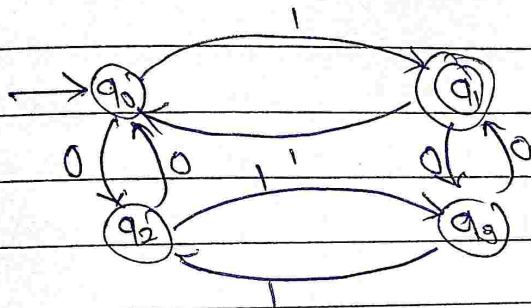


Q.2 Construct a DFA for the language over $(0,1)^*$ such that it contains "000" as a substring.

Ans. minimum string — 000. or $\Sigma = (0,1)$
construction

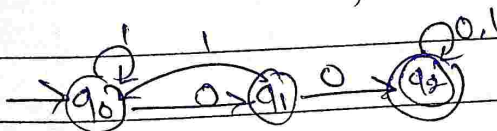


Q.3 Construct a DFA to recognize odd no. of 1's and even no. of 0's



q_0 — both even
 q_1 — 1 odd & 0 even.

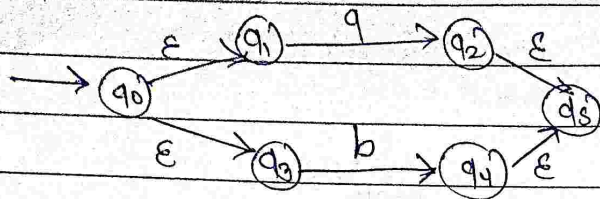
Q.4 Design DFA to accept strings over $\Sigma = (0,1)$ with "00"
minimum string : 00
 $\Sigma \neq (0,1)$.



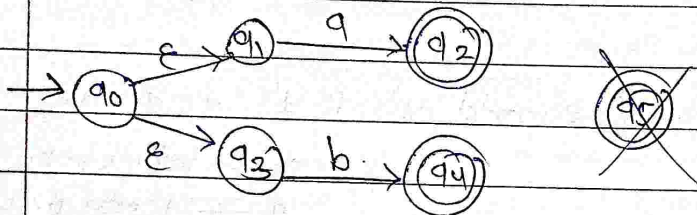
Q.5 Regex atleast one. $\{0,1\}^*$.
 $(0+1)^* . 1 . (0+1)^*$.

Q.9. NFA with ϵ -transitions.

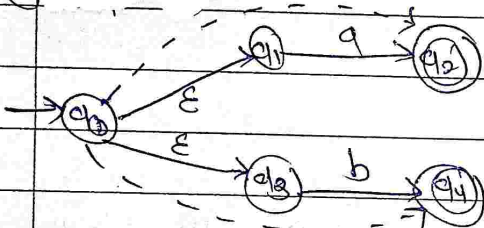
Step - 1



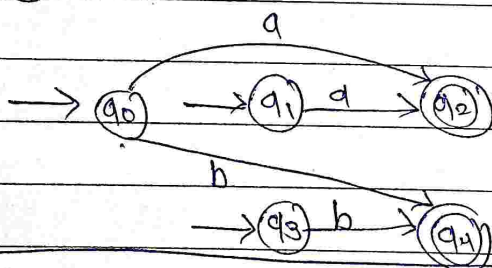
Step (2)



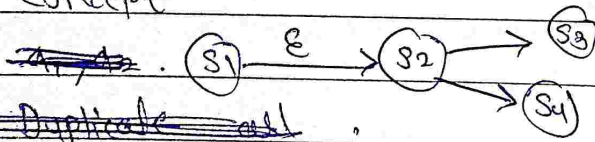
Step (3)



Step (4)



Concept.



- (1) Duplicate all edges going from s_2 in s_1 (duplicate in s_1)
- (2) If s_1 is initial then s_2 should be initial.
- (3) If s_2 is final then s_1 should be final.

Q.10. Convert NFA to DFA.

Ans. Step 1

current transition table.

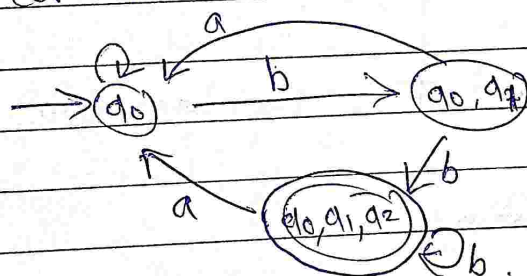
	a	b
q_0	q_0	$\{q_0, q_1\}$
q_1	—	q_2
q_2	—	—

Now, create a transition table of DFA with the help of above.

	a	b
Initial State. q_0	q_0	$\{q_0, q_1\}$
$\{q_0, q_1\}$	q_0	$\{q_0, q_1, q_2\}$
$\{q_0, q_1, q_2\}$	q_0	$\{q_0, q_1, q_2\}$

Final State

Now, construct DFA according to transition table.



Q.1.

DFA

NFA

①	one transition for one i/p	many transitions for one i/p
②	Doesn't allow ϵ (Epsilon)	allow.
③	One path for each i/p	many
④	More states.	Few.
⑤	Easy to Implement	Hard.
⑥	Can't convert to NFA	Can convert to DFA.
⑦	One path is followed by a string	Multiple
⑧	Deterministic finite Automata.	Non-deterministic finite automata

Q.6. Applications of finite Automata.

1) Compiler Design.

— used in lexical analyzers to identify tokens like keywords, identifiers
 ex. Detecting words like int, float, if in a program.

2) Pattern Matching.

Used to search patterns in text.

Ex. finding a word in a paragraph, (like search option).

3) Text editors.

Used in find & replace operations.

4) Spell checking.

used to check correct word pattern

4) Digital circuits.

used to design sequential circuits.
 Ex. Traffic light, Vending machine.

5) ATM machine.

Insert card
 Enter pin.
 Transaction.
 Withdraw.

$xyz \rightarrow y$ can be increase infinite times.

Page No.:

Date:

Q.8. State pumping lemma for regular languages.
Theorem.

It's used to design or prove that a language is not regular.
It can't be used to prove that language is regular. If A has a pumping length 'P' such that any string 's' where $|s| \geq P$ may be divided into three parts $s = xyz$. following conditions must be true.

- 1) $xyz \in A$ for every $i \geq 0$.
- 2) $|y| \geq 1$
- 3) $|xy| \leq P$.

Q.12. Explain Arden's theorem with example.

Ans. If we have a regular expression equation of the form.

$$R = Q + RP$$

where R, Q, P are regular expressions and P doesn't contain (ϵ) then eqn will be.

$$R = QP^*$$

or.

$$R = Q + RP \quad \text{--- (1) where } R = QP^*$$

put $r = QP^*$ in eqn (1).

$$R = Q + (QP^*P)$$

$$Q(\epsilon + P^*P)$$

$$QP^*$$

$$\dots \dots \dots [\epsilon + P^*P = P^*]$$

or.

put $R = Q + RP$ in eqn (1).

$$Q + [Q + RP]P$$

$$Q + QP + RP^2$$

$$Q + QP + (Q + RP)P^2$$

$$Q + QP + QP^2 + RP^3$$

$$Q + QP + QP^2 + \dots + QP^n + RP^{n+1} \quad [R = QP^*]$$

$$P + QP + QP^2 + \dots + QP + QP^{n+1}$$

$$Q[\epsilon + P + P^2 + \dots + P^n + P^{n+1}]$$

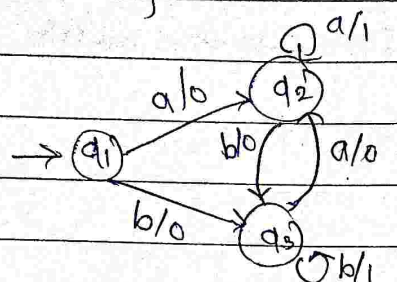
$$= QP^*$$

hence proved.

Mealy machine.

Design a mealy machine accepting the strings consisting of a and b from Σ^* , where $\Sigma = \{a, b\}$ and the strings should end with either aa or bb .

Ans min string aa or bb



Let's check it.

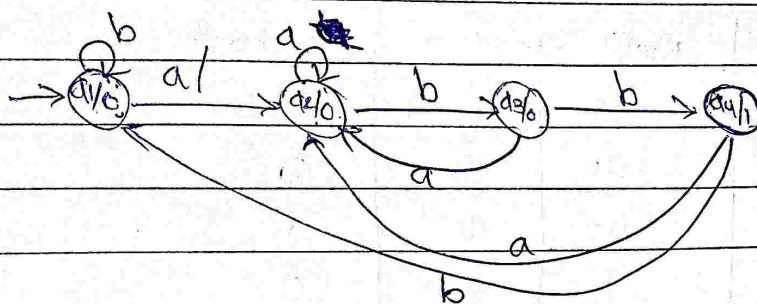
aabb.

q1	q2	q1	q3
0	1	0	1

Moore Machine.

Construct a Moore machine that counts the occurrences of the sequence 'abb' in any input strings over $\{a,b\}$.

Ans. $\Sigma = \{a,b\}$ $\Delta = \{0,1\}$
min string

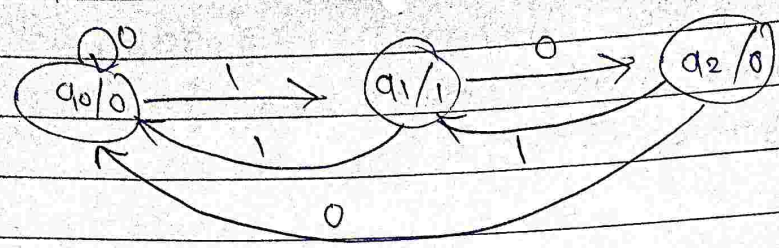


Let's check it.

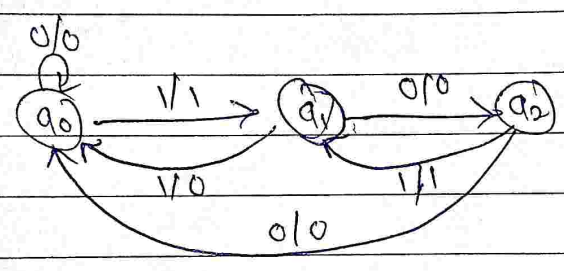
abb — string.

q_1	q_2	q_3	q_4
0	0	0	0

Q.14 Conversion of Moore to Mealy



Current State	Next state.		Output
	0	1	
q ₀	q ₀	q ₁	0
q ₁	q ₂	q ₀	1
q ₂	q ₀	q ₁	0



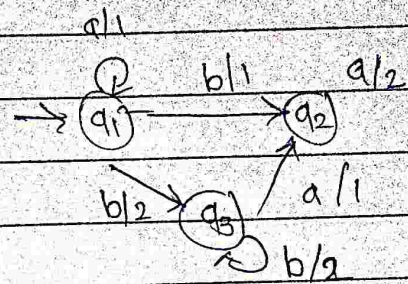
Mealy machine transition table.

Current state	Next state.		Next state.	
	0	1	0	1
q ₀	q ₀	q ₁	q ₀	q ₁
q ₁	q ₂	q ₀	q ₀	q ₁
q ₂	q ₀	q ₁	q ₁	q ₁

Q.15

Mealy to moore machine.

Current	Next	0	Next	1
	a		b	
q ₁	q ₁		q ₃	1
q ₂	q ₂		q ₁	1
q ₃	q ₂		q ₃	2

q₁ → 1q₂ → q₂ 1q₃ → q₃ 2

Moore machine transition table.

current	next state		output.
	a	b	
q ₁	q ₁	q ₃	1
q ₂	q ₂	q ₁	1
q ₃	q ₂	q ₃	2
q ₄	q ₂	q ₃	1
q ₅	q ₂	q ₃	2

